From Variants to Structure: Assessing Protein Structures of Hypertension-Associated Genes Using AlphaFold3 and TM-score/RMSD

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Abstract

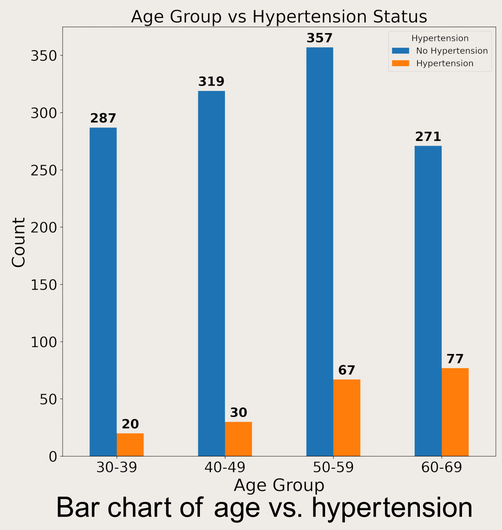
This study uses hypertension-associated genes from the GWAS Catalog and integrates them with Taiwan Biobank (TWB) data. Using an end-to-end pipeline, we convert VCFs[GRCh38/hg38] into haplotype(H1/H2) variant protein sequences and build the corresponding 3D models with AlphaFold3. By this process, we compared fixed and random seeds to found different, that a fixed seed produced more stable, better results, so we used a fixed seed for both the original sequences and variant sequences. We then performed global and local structural alignments between the two, using TM-score and RMSD to evaluate differences and explore how structural features relate to hypertension.

Motivation

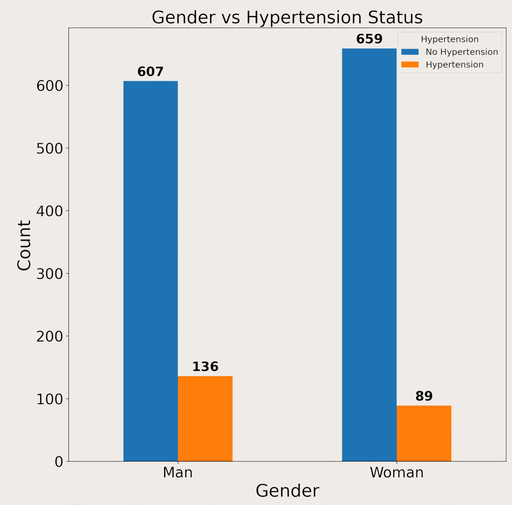
Genes determine a protein's sequence; the sequence shapes its structure; and structure drives function. GWAS has flagged many genes linked to hypertension, but what they actually do is often unclear. In this study, we use TWB VCF data to derive variant protein sequences, predict their 3D structures with AlphaFold3, and compare them globally and within specific domains to uncover structural features that may relate to hypertension.

Data Overview

From the TWB, we used VCF variant files, surveys (age, sex, self-reported hypertension, etc.), and physical exams (height, weight, blood pressure, etc.). We enrolled 1,495 participants; 4 couldn't be matched, so 1,491 sample were analyzed.



Bar chart of age vs. hypertension



Bar chart of sex vs. hypertension

AlphaFold2 VS. AlphaFold3

	AlphaFold2	AlphaFold3
Task	Predict single-protein 3D structures from sequence	All-atom complexes
Architecture	Evoformer + Structure Module (IPA)	Structure Module + Diffusion; Pairformer
Dependency and Training	Heavy MSA; performance drops when homologs are scarce	Cross-Distillation (leverages AF-Multimer)
Outputs	3D coords (monomer) + pLDDT, PAE	3D coords (all-atom complexes) + pLDDT, PAE, PDE

RMSE & TM-SCORE

$$\mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\|p_i - q_i\|^2}$$

N : number of aligned atoms
p_i, q_i : coordinates of atom i after optimal rotation and translation

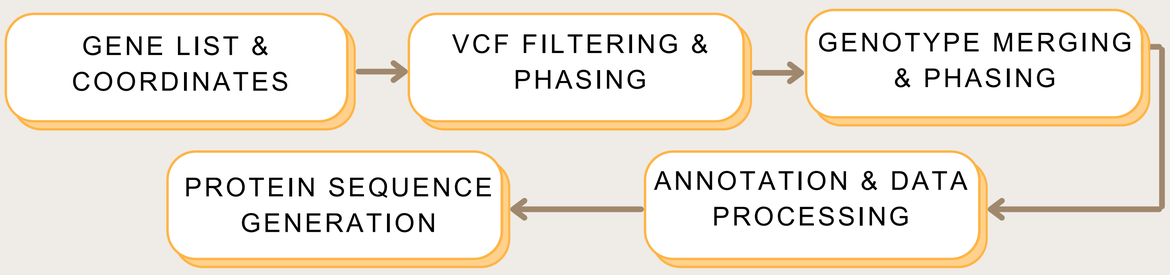
$$\text{TM-score} = \frac{1}{L_{\text{target}}}\sum_{i=1}^{L_{\text{aln}}}\frac{1}{1+\left(\frac{d_i}{d_0(L_{\text{target}})}\right)^2}$$

$$d_i = |\mathbf{x}'_i - \mathbf{y}_i|$$

$$d_0(L) = \max\left(0.5,\ 1.24(L-15)^{1/3} - 1.8\right)$$

$\mathbf{x}'_i$: coordinates of atom i in the aligned structure
$\mathbf{y}'_i$: coordinates of atom i in the reference structure

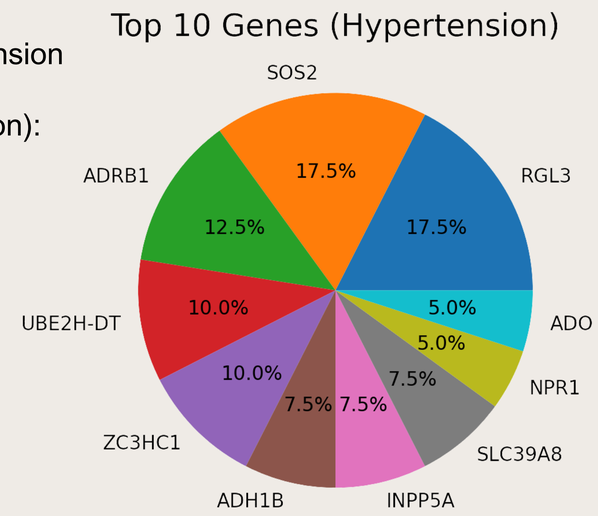
Gene-to-Protein Pipeline



Select Target Genes

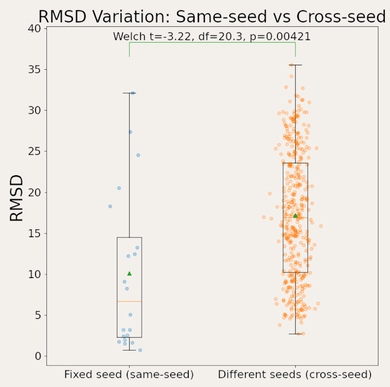
Selecting genes associated with hypertension

1. Disease keywords (strictly hypertension):
   - treatment-resistant hypertension
   - early-onset hypertension
   - pulmonary arterial hypertension
   - essential hypertension
   - hypertension
2. Significance threshold:
   - P-value < 5×10^{-8}
3. Filtering criterion:
   - CONTEXT is missense variant

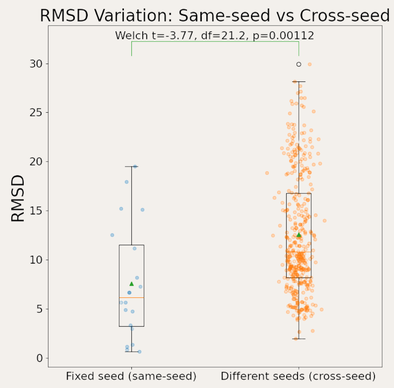


AlphaFold 3D Protein Structure Comparison

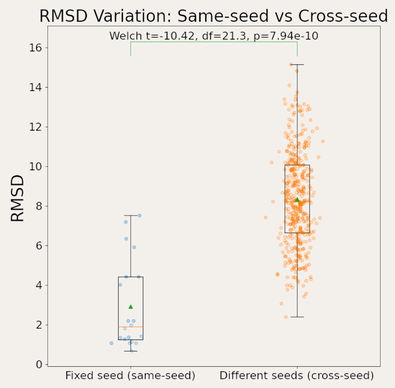
- Genes: **RGL3, ADRB1, NPR1** (GWAS hits associated with hypertension)
- Modeling choice: Repeated experiments and the boxplots show that using a fixed random seed in AlphaFold3 produces more consistent results than variable seeds, so we fix the seed for all subsequent modeling.



NPR1
Same seed vs Cross seed

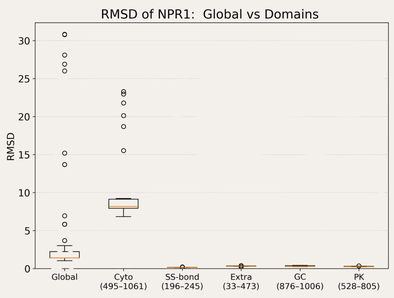


RGL3
Same seed vs Cross seed

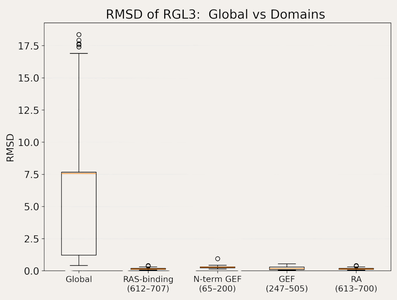


ADRB1
Same seed vs Cross seed

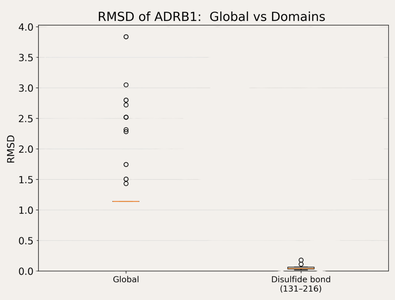
- Using both global and local structural alignments, we compared RGL3, ADRB1, and NPR1 and computed their RMSD and TM-score; the results are as follows.



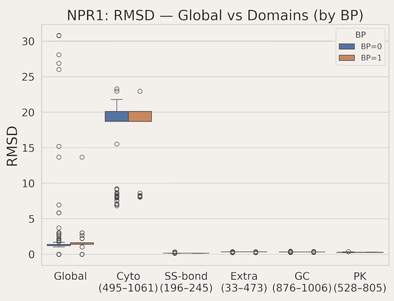
NPR1 Global vs Domains



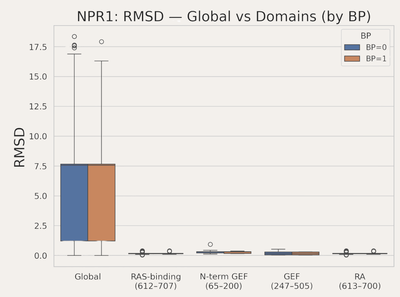
RGL3 Global vs Domains



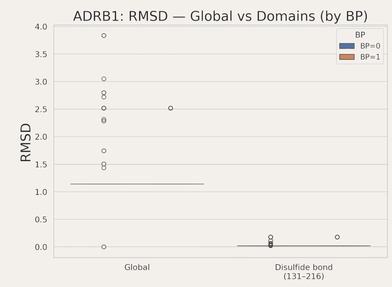
ADRB1 Global vs Domains



NPR1
Global vs Domains by Hypertension



RGL3
Global vs Domains by Hypertension



ADRB1
Global vs Domains by Hypertension

Conclusion and Disscussion

We built a **reproducible** end-to-end pipeline: starting from GWAS genes and TWB VCFs, we generated haplotype (H1/H2) variant protein sequences and modeled them with AlphaFold3. We focused on **missense variants**, RGL3, ADRB1, and NPR1. Comparing fixed vs. random seeds, t-tests gave p-values well below 0.05, that mean I have enough evidence to claim, so we used a fixed seed for all models. We then did global and local Cα alignments and evaluated RMSD and TM-score. RGL3 and NPR1 showed structural differences, but boxplots by hypertension status didn't separate clearly. At the domain level (UniProt), only the **NPR1 cytoplasmic** segment shifted, and it still didn't track with hypertension. Overall, with this gene set, structure hypertension links look limited. Because of AlphaFold's daily run limits, we couldn't scale further; next step is to include more genes and samples.

References

[1] Abramson, J., Adler, J., Dunger, J., et al. (2024). Accurate structure prediction of biomolecular interactions with AlphaFold 3. Nature, 630, 493–500. https://doi.org/10.1038/s41586-024-07487-w Nature
[2] Jumper, J., Evans, R., Pritzel, A., et al. (2021). Highly accurate protein structure prediction with AlphaFold. Nature, 596, 583–589. https://doi.org/10.1038/s41586-021-03819-2